

COMPREHENSIVE STATISTICS

A Complete Textbook Outline

57 chapters · 17 parts · 6 appendices

From probability foundations and descriptive statistics through Bayesian inference, causal methods, time series, machine learning, and epidemiology.

How to Read This Outline

This document is a master topic inventory, not a finished text. Each chapter heading corresponds to a self-contained unit; each numbered section is a major heading within that chapter; and each bullet is a concept, technique, or tool that requires exposition, worked examples, exercises, and figures in the final book.

Cross-references are embedded throughout (e.g., 'also Chapter 20') to flag where the same topic appears from a different perspective. The Appendix D decision flowchart and Appendix F discipline/software index are designed to be living documents updated as chapters are finalized.

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PART I

Foundations of Statistical Thinking

Chapter 1 The Nature of Statistics and Scientific Reasoning

1.1 What is statistics?

- Descriptive vs. inferential statistics
- The role of statistics in science and decision-making
- Historical development: from tables to machine learning
- Statistics, data science, and machine learning — relationships and distinctions

1.2 The statistical investigation cycle

- Problem formulation and research questions
- Study design
- Data collection
- Analysis
- Interpretation and communication
- Replication and cumulation

1.3 Populations, samples, and inference

- Population vs. sample
- Parameters vs. statistics
- Target population vs. study population
- Finite and infinite populations
- The inferential gap

1.4 Measurement and data types

- Nominal (categorical) scales
- Ordinal scales
- Interval scales
- Ratio scales
- Discrete vs. continuous variables
- Counts, proportions, rates, and ratios
- Latent vs. manifest variables

1.5 Sources of data

- Observational studies
- Experimental studies
- Administrative / found data
- Surveys and questionnaires
- Meta-data and big data

1.6 Causation and association

- Correlation does not imply causation
- Confounding and spurious association
- The potential-outcomes (counterfactual) framework
- Bradford Hill viewpoints
- When causal claims are warranted

1.7 Ethical considerations in statistics

- Data privacy and informed consent
- Fairness, bias, and algorithmic harm
- Reproducibility and research integrity
- Misuse of statistics

Chapter 2 Probability Theory

2.1 Foundations of probability

- Sample spaces and events
- Axiomatic foundations (Kolmogorov)
- Classical, frequentist, and subjective probability
- Logical / objective Bayesian probability

2.2 Set operations and event algebra

- Union, intersection, complement
- Mutually exclusive and exhaustive events
- De Morgan's laws

2.3 Rules of probability

- Addition rule (general and special case)
- Multiplication rule (general and special case)
- Conditional probability
- Independence of events
- Law of total probability
- Bayes' theorem and updating beliefs

2.4 Combinatorics and counting

- Permutations
- Combinations
- Multinomial coefficients
- Applications in probability calculations

2.5 Random variables

- Definition and notation
- Discrete random variables and probability mass functions
- Continuous random variables and probability density functions

- Cumulative distribution functions
- Quantile functions
- IID sequences, exchangeability, and de Finetti's representation theorem

2.6 Expectation and variance

- Expected value and its properties
- Variance and standard deviation
- Covariance and correlation
- Moment-generating functions
- Characteristic functions

2.7 Functions of random variables

- Transformations of single variables
- Sums of random variables
- Order statistics
- The delta method

2.8 Limit theorems

- Law of large numbers (weak and strong)
- Central limit theorem
- Berry-Esseen theorem and rates of convergence
- Continuous mapping theorem
- Slutsky's theorem
- Le Cam's theorem (Poisson approximation for rare events)
- Distances between distributions: total variation, Kullback-Leibler, Hellinger; Pinsker's inequality

Chapter 3 Probability Distributions

3.1 Families and their properties

- Location, scale, and shape parameters
- Exponential family of distributions
- Heavy tails vs. light tails
- Skewness and kurtosis across families

3.2 Discrete distributions

- Bernoulli and binomial
- Poisson
- Negative binomial and geometric
- Hypergeometric
- Multinomial
- Zero-inflated and zero-truncated variants

3.3 Continuous distributions

- Uniform

- Normal (Gaussian)
- Log-normal
- Exponential
- Gamma and chi-squared
- Beta
- Weibull
- Cauchy and heavy-tailed distributions
- Laplace
- Student's t, F, and their genesis

3.4 Multivariate distributions

- Multivariate normal
- Multivariate t
- Wishart and inverse-Wishart
- Dirichlet
- Copulas and dependence structures

3.5 Sampling distributions

- Distribution of the sample mean
- Distribution of the sample variance
- Chi-squared, t, and F as sampling distributions
- Exact vs. approximate sampling distributions

3.6 Extreme-value distributions

- Generalized extreme value (GEV) and its three types
- Generalized Pareto distribution
- Extreme-value theory in brief

3.7 Distribution fitting and selection

- QQ plots and PP plots
- Empirical CDF
- Goodness-of-fit testing (overview — details in Chapter 11)
- AIC/BIC for distribution selection

Chapter 4 Descriptive Statistics and Data Visualization

4.1 Measures of central tendency

- Arithmetic mean
- Geometric and harmonic mean
- Median
- Mode
- Trimmed and Winsorized means
- Weighted mean

4.2 Measures of dispersion

- Range and interquartile range
- Variance and standard deviation
- Coefficient of variation
- Mean absolute deviation
- Gini coefficient

4.3 Measures of shape

- Skewness: definition, moments, Pearson and Bowley measures
- Kurtosis: excess kurtosis, leptokurtic and platykurtic distributions
- L-moments
- Interpreting departures from normality

4.4 Quantiles, order statistics, and the five-number summary

- Percentiles and quantiles
- Order statistics and their distributions
- Boxplots and their variants

4.5 Data visualization principles

- Grammar of graphics
- Histograms and density plots
- Stem-and-leaf plots
- Boxplots
- Scatterplots and scatterplot matrices
- Bar and dot charts
- Heat maps and mosaic plots
- Visualization for multivariate data
- Common visualization mistakes

4.6 Exploratory data analysis (EDA)

- Tukey's EDA philosophy
- Resistant statistics
- Transformations in EDA
- Identifying patterns, clusters, and anomalies

4.7 Summary tables and cross-tabulations

- Frequency and relative-frequency tables
- Contingency tables
- Pivot tables

PART II

Collecting Data: Sampling and Experimental Design

Chapter 5 Sampling Theory and Survey Methods

5.1 Sampling principles

- Simple random sampling
- Systematic sampling
- Stratified sampling
- Cluster sampling
- Multistage sampling
- Probability-proportional-to-size sampling

5.2 Non-probability sampling

- Convenience, quota, purposive, and snowball sampling
- When non-probability sampling is acceptable

5.3 Estimation under complex designs

- Design-based vs. model-based inference
- Horvitz-Thompson estimator
- Design effects and effective sample size
- Post-stratification and calibration
- Replicate weights (jackknife, BRR, bootstrap)

5.4 Bias and error in surveys

- Sampling error vs. non-sampling error
- Coverage error
- Non-response bias and adjustments
- Measurement error and social-desirability bias
- Mode effects

5.5 Special survey techniques

- Randomized response
- Capture-recapture methods
- Network and respondent-driven sampling

Chapter 6 Experimental Design

6.1 Principles of experimentation

- Randomization
- Replication
- Blocking
- Control

6.2 Basic designs

- Completely randomized design
- Randomized complete block design
- Latin square design
- Graeco-Latin square

6.3 Factorial designs

- Full factorial
- Fractional factorial and aliasing
- Confounding in factorial designs
- 2^k designs
- Resolution and defining relations

6.4 Advanced designs

- Split-plot design
- Strip-plot design
- Nested designs
- Crossover designs and carryover effects
- Incomplete block designs (BIBD, PBIBD)

6.5 Response surface methodology

- First- and second-order designs
- Central composite and Box-Behnken designs
- Optimization and canonical analysis

6.6 Taguchi and robust design

- Signal-to-noise ratios
- Orthogonal arrays
- Parameter design

6.7 Sequential and adaptive designs

- Sequential experimentation
- Optimal stopping
- Group sequential and adaptive clinical trial designs

6.8 Sample-size planning

- Power and Type II error
- Sample-size formulas for common designs
- Simulation-based power
- Design effects for clustered experiments

PART III**Statistical Inference: Foundations**

Chapter 7 Estimation Theory**7.1 Point estimation**

- Estimators and their properties
- Unbiasedness
- Consistency
- Efficiency and the Cramér-Rao lower bound
- Sufficiency and completeness

7.2 Estimation methods

- Maximum likelihood estimation (MLE)
- Method of moments
- Least squares
- Generalized method of moments (GMM)
- M-estimation
- Minimum distance estimation
- Bayesian estimation (MAP, posterior mean)

7.3 Interval estimation

- Confidence intervals: definition and interpretation
- Exact vs. approximate CIs
- Wald, score, and likelihood-ratio CIs
- Profile-likelihood CIs
- Bootstrap CIs (percentile, BCa, studentized)
- CIs for proportions (Wald, Wilson, Clopper-Pearson, Agresti-Coull)
- Fieller's theorem for ratios
- Delta method

7.4 Shrinkage and regularized estimation

- James-Stein and empirical Bayes
- Ridge, lasso, and elastic net as estimators
- Bias-variance trade-off

Chapter 8 Hypothesis Testing: Foundations and Philosophy**8.1 The logic of hypothesis testing**

- Null and alternative hypotheses
- One-sided vs. two-sided tests
- The significance level α

- Rejection regions and critical values

8.2 Error types and operating characteristics

- Type I error (α) and Type II error (β)
- Type III error (wrong direction)
- Type IV error (misinterpreting a rejection)
- Type S (sign) and Type M (magnitude) errors — Gelman-Carlin framework
- Power ($1 - \beta$) and its determinants
- Power curves
- Operating characteristic curves

8.3 The p-value: meaning and misuse

- Definition of the p-value
- What the p-value is not
- The p-value under the alternative
- ASA statement on p-values
- Alternatives: Bayes factors, e-values

8.4 General test construction

- Neyman-Pearson lemma and uniformly most powerful tests
- Likelihood-ratio tests
- Wald tests
- Score (Lagrange multiplier) tests
- The trinity of classical tests

8.5 Multiple testing

- The multiple-comparisons problem
- Family-wise error rate (FWER): Bonferroni, Holm, Hochberg, Hommel, Šidák
- False discovery rate (FDR): Benjamini-Hochberg, Benjamini-Yekutieli
- Simultaneous inference
- Pre-registration and reducing implicit tests

8.6 Philosophical debates in inference

- Frequentist vs. Bayesian vs. likelihood approaches
- The replication crisis and its causes
- Pre-registration and registered reports
- The 'new statistics': effect sizes, CIs, and meta-analysis
- Equivalence testing and TOST
- Moving beyond $p < 0.05$

8.7 Sequential testing and always-valid inference

- The problem with optional stopping
- Sequential probability ratio test (SPRT)
- E-values and anytime-valid p-values
- Group sequential designs
- Alpha-spending functions

PART IV

Inference for Means and Proportions

Chapter 9 One-Sample and Two-Sample Inference

9.1 Inference for a single mean

- One-sample z-test (known σ)
- One-sample t-test
- Confidence intervals for a mean
- Sample-size determination

9.2 Inference for two means

- Independent two-sample t-test (equal variances)
- Welch's t-test (unequal variances)
- Paired-samples t-test
- Choosing between paired and independent designs
- Equivalence tests for two means (TOST)

9.3 Inference for a single proportion

- Binomial test
- One-proportion z-test
- Confidence intervals: Wald, Wilson, Clopper-Pearson, Agresti-Coull

9.4 Inference for two proportions

- Two-proportion z-test
- Fisher's exact test and Barnard's test / Boschloo's test
- McNemar's test for paired binary data
- Risk difference, relative risk, and odds ratio
- Confidence intervals for a difference of proportions: Wald and Agresti-Caffo

9.5 Inference for variances

- One-sample chi-squared test for variance
- F-test for two variances
- Robustness and alternatives

9.6 Non-parametric alternatives

- Sign test
- Wilcoxon signed-rank test
- Mann-Whitney U / Wilcoxon rank-sum test
- Brunner-Munzel test
- Choosing parametric vs. non-parametric

Chapter 10 Categorical Data Analysis

10.1 Contingency tables and association

- Chi-squared test of independence
- G-test (likelihood-ratio chi-squared)
- Fisher's exact test (and extensions)
- Barnard's test and Boschloo's test
- Measures: phi, Cramér's V, Goodman-Kruskal lambda and gamma, Yule's Q

10.2 Goodness-of-fit

- Chi-squared goodness-of-fit
- G-test goodness-of-fit
- Multinomial test

10.3 Stratified and matched data

- Cochran-Mantel-Haenszel test
- Breslow-Day test of interaction
- Mantel-Haenszel pooled odds-ratio estimator; stratification as the resolution of Simpson's paradox (Chapter 45)
- McNemar's test
- Cochran's Q test

10.4 Trend tests

- Cochran-Armitage trend test
- Jonckheere-Terpstra test

10.5 Ordinal data

- Proportional-odds model (overview — details in Chapter 14)
- Spearman and Kendall correlation for ordinal data

Chapter 11 Distribution Testing and Assumption Checking

11.1 Testing for normality

- Shapiro-Wilk test
- Shapiro-Francia test
- Kolmogorov-Smirnov and Lilliefors tests
- Anderson-Darling test
- Cramér-von Mises test
- Jarque-Bera and D'Agostino K^2 tests
- Ryan-Joiner test
- QQ plots and their interpretation

11.2 Testing multivariate normality

- Mardia's test
- Henze-Zirkler test
- Royston's test
- Multivariate QQ plots

11.3 Testing for homogeneity of variance

- Levene's test
- Brown-Forsythe test
- Bartlett's test
- Fligner-Killeen test
- Hartley's F-max test
- Siegel-Tukey, Ansari-Bradley, and Mood's scale tests

11.4 General goodness-of-fit

- Kolmogorov-Smirnov (two-sample and one-sample)
- Anderson-Darling
- Chi-squared for binned data
- Cramér-von Mises

11.5 Testing sphericity and covariance structure

- Mauchly's test
- Greenhouse-Geisser and Huynh-Feldt corrections
- Box's M test
- Bartlett's test of sphericity (factor analysis)

11.6 Outlier and influence detection

- Grubbs' test
- Dixon's Q test
- Rosner's ESD (generalized extreme studentized deviate)
- Tietjen-Moore test
- Cook's distance, leverage, and DFFITS

PART V

Comparing Multiple Groups: ANOVA and Extensions

Chapter 12 Analysis of Variance

12.1 One-way ANOVA

- Partitioning of sums of squares
- The F-test and its rationale
- Assumptions: normality, homoscedasticity, independence
- Effect sizes: η^2 , partial η^2 , ω^2
- Power and sample-size planning
- Welch's ANOVA for unequal variances

12.2 Post-hoc and multiple comparison procedures

- FWER control: Tukey HSD, Bonferroni, Holm, Hochberg, Hommel, Šidák, Scheffé, Dunnett
- FDR control: Benjamini-Hochberg, Benjamini-Yekutieli
- Non-parametric post-hoc: Dunn, Nemenyi
- Planned contrasts vs. post-hoc comparisons

12.3 Factorial ANOVA

- Two-way ANOVA: main effects and interactions
- Higher-order factorial designs
- Unbalanced designs and Type I/II/III sums of squares
- Simple main effects analysis

12.4 ANCOVA

- Rationale and assumptions
- Homogeneity of regression slopes
- Adjusted means
- ANCOVA vs. stratified analysis

12.5 Repeated-measures ANOVA

- One-way repeated measures
- Sphericity and corrections
- Mixed-design (split-plot) ANOVA
- Multivariate approach to repeated measures

12.6 Non-parametric alternatives

- Kruskal-Wallis test
- Friedman test
- Page's trend test

- Mood's median test

Chapter 13 Multivariate Analysis of Variance

13.1 MANOVA foundations

- Multivariate hypotheses
- Wilks' lambda, Pillai's trace, Hotelling-Lawley trace, Roy's largest root
- Assumptions and robustness
- Follow-up procedures: discriminant analysis, univariate ANOVAs

13.2 MANCOVA

- Multivariate analysis of covariance
- Assumptions and interpretation

13.3 Hotelling's T^2

- One-sample and two-sample multivariate tests

13.4 Profile analysis

- Parallelism, levels, and flatness tests

PART VI

Regression and the Generalized Linear Model

Chapter 14 Simple and Multiple Linear Regression

14.1 Simple linear regression

- The model and assumptions
- OLS estimation
- Inference for slope and intercept
- Prediction intervals vs. confidence intervals for the mean
- R^2 and its interpretation

14.2 Multiple linear regression

- Matrix formulation
- Partial slopes and their interpretation
- Frisch-Waugh-Lovell theorem: partialling out via the residual-maker (annihilator) matrix
- Adjusted R^2 and model fit
- Multicollinearity and VIF
- Standardized coefficients

14.3 Model building and variable selection

- Nested model comparison: F-test and LRT
- AIC, BIC, AICc, Mallows's C_p
- Stepwise, forward, and backward selection (and their limits)
- All-subsets regression
- Cross-validation for model selection

14.4 Regression diagnostics

- Residual plots and their interpretation
- Testing heteroscedasticity: Breusch-Pagan, White, Goldfeld-Quandt
- Testing for autocorrelation: Durbin-Watson, Breusch-Godfrey
- Ramsey RESET test for misspecification
- Harvey-Collier test for linearity
- Influential observations: leverage, Cook's distance, DFFITS
- Added-variable (partial regression) plots

14.5 Remedial measures

- Weighted least squares (WLS) and GLS
- Sandwich (Huber-White) standard errors
- Cluster-robust standard errors
- Transformations (see Chapter 20 for detail)
- Polynomial and interaction terms

Chapter 15 The Generalized Linear Model

15.1 GLM framework

- The three components: random, systematic, link
- Exponential family and canonical links
- Deviance and likelihood-ratio tests in GLMs
- Residuals: Pearson, deviance, Anscombe
- Overdispersion and quasi-likelihood

15.2 Logistic regression

- Binary outcomes and the logit link
- Interpretation of odds ratios
- Hosmer-Lemeshow goodness-of-fit
- ROC/AUC and classification performance
- Separation and rare events

15.3 Extensions of logistic regression

- Multinomial logistic regression
- Ordinal (proportional-odds) regression
- Stereotype logit model
- Probit, complementary log-log, and cauchit links

15.4 Poisson regression and count models

- Poisson regression
- Negative binomial regression
- Quasi-Poisson
- Zero-inflated models (ZIP, ZINB)
- Zero-truncated models
- Hurdle models

15.5 Other GLM families

- Gamma regression
- Inverse-Gaussian regression
- Tweedie (compound Poisson-gamma) regression: power mean-variance family for zero-inflated continuous outcomes
- Tobit regression (censored outcomes)
- Beta regression (proportions)

Chapter 16 Regularized, Flexible, and Nonparametric Regression

16.1 Regularized regression

- Ridge regression
- Lasso regression and variable selection

- Elastic net
- Adaptive lasso
- Choosing the regularization parameter

16.2 Polynomial and spline regression

- Polynomial regression
- Natural and B-splines
- Smoothing splines
- Penalized splines (P-splines)
- Selecting knots and smoothness

16.3 Local and kernel regression

- LOESS / LOWESS
- Kernel regression (Nadaraya-Watson)
- Bandwidth selection

16.4 Generalized additive models (GAMs)

- Additive model framework
- Backfitting and PIRLS
- GAMMs (with random effects)
- Tensor product smooths

16.5 Quantile regression

- Median regression and beyond
- Inference and interpretation
- Quantile regression forests

16.6 Gaussian process regression

- Gaussian process prior
- Kernel functions
- Prediction and uncertainty

Chapter 17 Mixed-Effects Models and Correlated Data

17.1 Why mixed models?

- Clustered and hierarchical data
- Repeated measures and longitudinal data
- Random effects and their interpretation

17.2 Linear mixed-effects models (LMM)

- Random intercepts and slopes
- REML estimation
- Inference: likelihood ratio, parametric bootstrap
- Intraclass correlation

17.3 Generalized linear mixed models (GLMM)

- Binary, count, and ordinal outcomes with clustering
- Approximation methods: PQL, Laplace, AGH
- Subject-specific vs. population-averaged interpretation

17.4 Generalized estimating equations (GEE)

- Working correlation structures
- Robust standard errors
- GEE vs. GLMM

17.5 Growth-curve and trajectory models

- Polynomial growth curves
- Piecewise / change-point models
- Latent growth curve models in SEM

PART VII

Correlation, Association, and Multivariate Methods

Chapter 18 Correlation and Measures of Association

18.1 Bivariate correlation

- Pearson r
- Spearman ρ and Kendall τ
- Point-biserial and phi correlations
- Partial and semi-partial correlation
- Intraclass correlation (ICC forms)

18.2 Measures of association for categorical data

- Phi, Cramér's V , Goodman-Kruskal gamma and lambda
- Kendall's tau-b and tau-c for ordinal tables
- Uncertainty coefficient

18.3 Advanced dependence measures

- Distance correlation
- Mutual information
- Copulas
- Maximal information coefficient (MIC)

18.4 Agreement and reliability

- Cohen's kappa, weighted kappa, Fleiss' kappa
- Krippendorff's alpha
- Cronbach's alpha and McDonald's omega
- Kuder-Richardson KR-20
- Kendall's W
- Bland-Altman analysis
- ICC in reliability contexts

Chapter 19 Dimension Reduction and Latent-Structure Methods

19.1 Principal component analysis (PCA)

- Eigendecomposition and singular value decomposition
- Choosing the number of components
- Biplots and loadings
- Robust PCA

19.2 Factor analysis

- Exploratory factor analysis (EFA): extraction and rotation
- Confirmatory factor analysis (CFA)
- Identification and model fit indices
- EFA vs. PCA

19.3 Independent component analysis (ICA)

- Non-Gaussianity and independence
- FastICA and infomax algorithms
- Applications: blind source separation, fMRI

19.4 Canonical correlation analysis (CCA)

- Maximally correlated linear combinations
- Regularized CCA and kernel CCA

19.5 Multidimensional scaling (MDS)

- Classical MDS
- Non-metric MDS
- Procrustes analysis

19.6 Correspondence analysis

- Simple and multiple correspondence analysis
- Biplot interpretation

19.7 Manifold learning and nonlinear methods

- t-SNE
- UMAP
- Isomap and locally linear embedding

19.8 Partial least squares (PLS) and related

- PLS regression and PLS-DA
- Canonical partial least squares

Chapter 20 Cluster Analysis

20.1 Distance and similarity measures

- Euclidean, Manhattan, Minkowski
- Mahalanobis distance
- Cosine similarity
- Distances for categorical and mixed data

20.2 Hierarchical clustering

- Agglomerative methods: single, complete, average, Ward

- Divisive clustering
- Dendrograms and cutting

20.3 Partitional clustering

- k-means and k-medoids (PAM)
- Choosing k: elbow, silhouette, gap statistic
- k-means++ initialization

20.4 Model-based clustering

- Gaussian mixture models
- EM algorithm
- BIC for model selection

20.5 Density-based clustering

- DBSCAN
- HDBSCAN

20.6 Cluster validation

- Internal indices: silhouette, Calinski-Harabasz, Davies-Bouldin
- External indices: Rand index, adjusted Rand index
- Stability-based validation

Chapter 21 Discriminant Analysis and Classification

21.1 Linear discriminant analysis (LDA)

- Fisher's discriminant criterion
- Assumptions and robustness
- Classification rule and posterior probabilities

21.2 Quadratic discriminant analysis (QDA)

- When covariance matrices differ

21.3 Regularized discriminant analysis

- Shrinkage estimators for covariance

21.4 Naive Bayes classification

- Independence assumption
- Gaussian, Bernoulli, and multinomial Naive Bayes

PART VIII

Structural Equation, Causal, and Mediation Models

Chapter 22 Structural Equation Modeling (SEM)

22.1 Path analysis

- Path diagrams and coefficients
- Direct and indirect effects
- Path tracing rules

22.2 SEM framework

- Measurement model (CFA)
- Structural model (path model)
- Identification conditions
- Estimation: ML, GLS, WLS, DWLS

22.3 Model fit and comparison

- Chi-squared test of exact fit
- RMSEA, CFI, TLI, SRMR
- Modification indices and their risks
- Comparing nested and non-nested SEMs

22.4 Mediation and moderation in SEM

- Baron-Kenny causal-steps approach
- Sobel test
- Bootstrapped indirect effects
- Moderated mediation and mediated moderation
- PROCESS macro and lavaan

22.5 Latent growth curve models

- Growth trajectory parameters
- Predictors of growth

22.6 Multi-group SEM

- Measurement invariance testing
- Configural, metric, and scalar invariance

Chapter 23 Causal Inference

23.1 Frameworks for causality

- Potential outcomes / Rubin causal model
- Directed acyclic graphs (DAGs) and do-calculus
- Structural causal models (Pearl)
- Assumptions: SUTVA, positivity, ignorability

23.2 Experimental approaches

- Randomized controlled trials
- Encouragement designs and partial compliance
- Factorial experiments for causal effects

23.3 Observational approaches

- Propensity score matching and weighting (IPW, AIPW)
- Stratification and covariate adjustment
- Doubly robust estimators

23.4 Natural experiments

- Instrumental variables (IV) and two-stage least squares
- Difference-in-differences (DiD) and parallel trends
- Regression discontinuity design (sharp and fuzzy RDD)
- Synthetic control method
- Interrupted time series

23.5 Sensitivity analysis for unmeasured confounding

- Rosenbaum bounds
- E-values
- Quantitative bias analysis

23.6 Mendelian randomization

- Using genetic variants as instruments
- Two-sample MR
- Sensitivity analyses: MR-Egger, weighted median

PART IX

Survival Analysis and Longitudinal Methods

Chapter 24 Survival Analysis

24.1 Survival data fundamentals

- Survival function, hazard function, cumulative hazard
- Types of censoring (right, left, interval) and truncation
- Person-time and incidence rates

24.2 Non-parametric methods

- Kaplan-Meier estimator
- Nelson-Aalen estimator
- Log-rank, Gehan-Breslow, Tarone-Ware tests
- Restricted mean survival time (RMST)

24.3 Cox proportional-hazards model

- Partial likelihood
- Tied failures
- Testing the PH assumption: Schoenfeld residuals, log-log plots
- Time-varying covariates
- Stratified Cox model

24.4 Parametric survival models

- Exponential, Weibull, Gompertz, log-normal, log-logistic
- Accelerated failure time (AFT) framework

24.5 Extensions

- Competing-risks models (cause-specific hazards, Fine-Gray)
- Frailty models (shared and individual)
- Cure models
- Multi-state models

Chapter 25 Longitudinal and Panel Data Analysis

25.1 Longitudinal data structure

- Balanced vs. unbalanced panels
- Wide vs. long format
- Missing data patterns in longitudinal studies

25.2 Panel data models (econometric)

- Pooled OLS
- Fixed-effects model and within estimator
- Random-effects model and GLS
- Hausman test
- First-difference estimator

25.3 Dynamic panel models

- Lagged dependent variable
- Arellano-Bond (GMM) estimator
- Arellano-Bover / Blundell-Bond system GMM

25.4 Panel unit roots and cointegration

- Levin-Lin-Chu, Im-Pesaran-Shin, Fisher-type tests
- Pedroni and Kao cointegration tests

PART X

Time Series Analysis

Chapter 26 Time Series: Foundations and Classical Models

26.1 Time series concepts

- Stationarity (weak and strict)
- Autocovariance and autocorrelation functions (ACF, PACF)
- White noise and random walks
- Decomposition: trend, seasonality, cycle, irregular

26.2 Stationarity testing

- Augmented Dickey-Fuller (ADF)
- Phillips-Perron test
- KPSS test (reversed null)
- Zivot-Andrews test (with structural break)

26.3 ARIMA models

- AR, MA, and ARMA processes
- Integration and differencing
- Box-Jenkins identification, estimation, and diagnostic checking
- Ljung-Box and Box-Pierce tests
- Seasonal ARIMA (SARIMA)

26.4 Structural break tests

- Chow test
- CUSUM and CUSUM-of-squares tests
- Bai-Perron multiple break test

26.5 Cointegration and error correction

- Engle-Granger two-step method
- Johansen trace and max-eigenvalue tests
- Vector error correction model (VECM)

Chapter 27 Advanced Time Series and Forecasting

27.1 Multivariate time series

- VAR (vector autoregression)
- Granger causality test
- Variance-ratio test (Lo-MacKinlay)

27.2 Volatility models

- ARCH and GARCH models
- Engle's ARCH test
- Asymmetric GARCH (EGARCH, GJR-GARCH)
- Stochastic volatility models

27.3 State-space models and the Kalman filter

- State-space representation
- Kalman filter and smoother
- Structural time series models

27.4 Spectral analysis

- Periodogram
- Spectral density estimation
- Fourier and wavelet transforms

27.5 Forecasting

- Exponential smoothing (ETS models)
- Holt-Winters models
- Diebold-Mariano forecast comparison test
- Forecast evaluation: MAE, RMSE, MAPE, CRPS

27.6 Non-linear and modern time series

- Threshold AR (TAR/SETAR)
- MARKOV switching models
- Neural network time series (LSTM, temporal convolutional networks)
- Prophet and other automated forecasting

PART XI

Bayesian Statistics

Chapter 28 Bayesian Inference

28.1 Bayesian philosophy and foundations

- Subjective vs. objective Bayesian approaches
- Prior, likelihood, and posterior
- Bayes' theorem for continuous parameters
- Conjugate priors
- Jeffreys and other non-informative priors

28.2 Bayesian estimation

- Posterior mean, median, and MAP
- Credible (highest-density) intervals
- Prior sensitivity analysis

28.3 Bayesian hypothesis testing

- Bayes factors
- BIC approximation to Bayes factors
- Bayesian t-tests (BEST)
- Posterior predictive p-values

28.4 MCMC methods

- Metropolis-Hastings algorithm
- Gibbs sampler
- Hamiltonian Monte Carlo (HMC) and NUTS
- Gelman-Rubin convergence diagnostic (R-hat)
- Effective sample size (ESS)
- Trace plots and diagnostics

28.5 Variational inference

- Mean-field variational Bayes
- ELBO and KL divergence
- Expectation propagation

28.6 Bayesian model selection and averaging

- DIC, WAIC, LOO-CV
- Bayesian model averaging (BMA)
- Spike-and-slab priors for variable selection

28.7 Bayesian hierarchical models

- Partial pooling and shrinkage
- Hyperpriors
- Connection to mixed-effects models

Chapter 29 Bayesian Computation and Applications

29.1 Software ecosystems

- Stan and brms
- PyMC and NumPyro
- JAGS and BUGS

29.2 Approximate Bayesian computation (ABC)

- Likelihood-free inference
- Summary statistics and tolerance

29.3 Sequential Monte Carlo (SMC / particle filters)

- Online Bayesian updating
- Applications in tracking and state estimation

29.4 Gaussian processes (Bayesian view)

- GP prior and posterior
- Kernel design
- Bayesian optimization

PART XII

Resampling, Robust, and Non-parametric Methods

Chapter 30 Resampling Methods

30.1 Bootstrap

- Non-parametric bootstrap: theory and practice
- Parametric bootstrap
- Bootstrap confidence intervals: percentile, basic, BCa, studentized
- Bootstrap hypothesis tests
- Double bootstrap

30.2 Jackknife

- Leave-one-out jackknife
- Bias and variance estimation
- Delete-d jackknife

30.3 Permutation and randomization tests

- Theory and exchangeability
- Exact permutation tests
- Monte Carlo permutation
- PERMANOVA

30.4 Cross-validation

- k-fold and LOOCV
- Nested cross-validation
- Repeated and stratified CV
- Bootstrap vs. cross-validation
- .632 estimator

Chapter 31 Robust Statistics

31.1 Concepts of robustness

- Breakdown point
- Influence function
- Sensitivity curve

31.2 Robust location and scale

- Trimmed and Winsorized means

- Hodges-Lehmann estimator
- Median absolute deviation (MAD)
- Rousseeuw-Croux scale estimators

31.3 Robust regression

- M-estimation (Huber, Tukey bisquare)
- MM-estimation
- Least trimmed squares (LTS)
- Theil-Sen estimator
- RANSAC

31.4 Robust inference

- Sandwich (Huber-White) standard errors
- HC0–HC5 corrections
- Cluster-robust and panel-robust SEs
- Robust Wald and score tests

Chapter 32 Comprehensive Non-parametric Methods

32.1 One- and two-sample rank tests

- Sign test
- Wilcoxon signed-rank test
- Mann-Whitney U / Wilcoxon rank-sum
- Brunner-Munzel test

32.2 k-sample rank tests

- Kruskal-Wallis
- Friedman
- Page's trend test
- Cochran's Q

32.3 Scale and dispersion tests

- Siegel-Tukey, Ansari-Bradley, Mood's scale test
- Fligner-Killeen

32.4 Runs, randomness, and independence

- Wald-Wolfowitz runs test
- Turning-point test

32.5 Density and CDF estimation

- Empirical CDF and DKW inequality
- Kernel density estimation
- Bandwidth selection methods

PART XIII**Specialized Topics and Domain Methods**

Chapter 33 Missing Data**33.1 Mechanisms of missingness**

- MCAR, MAR, and MNAR
- Little's MCAR test
- Identifying the mechanism in practice

33.2 Simple and ad-hoc approaches

- Complete-case analysis
- Mean and single imputation
- Last observation carried forward (LOCF)
- When they are and are not appropriate

33.3 Multiple imputation

- Multiple imputation by chained equations (MICE)
- Rubin's rules for pooling
- Imputation models and predictor selection
- Diagnostics

33.4 Model-based approaches

- Expectation-maximization (EM) algorithm
- Full-information maximum likelihood (FIML)
- Bayesian imputation

33.5 Weighting approaches

- Inverse-probability weighting (IPW)
- Doubly robust estimation

Chapter 34 Effect Sizes and Statistical Power**34.1 Why effect sizes?**

- Limitations of p-values alone
- Practical vs. statistical significance
- Effect size in the replication crisis

34.2 Standardized mean differences

- Cohen's d, Hedges' g, Glass's Δ

- Small/medium/large benchmarks and their critiques

34.3 Variance-explained measures

- η^2 , partial η^2 , ω^2 , ε^2
- R^2 , adjusted R^2 , Cohen's f and f^2

34.4 Association and risk measures

- Phi, Cramér's V , Odds ratio, Risk ratio, Hazard ratio
- Cliff's delta and rank-biserial r
- Number needed to treat / harm (NNT/NNH)

34.5 Statistical power

- A priori power analysis
- Post-hoc power and its problems
- Sensitivity analysis: MDE for fixed n
- Simulation-based power
- Power for complex designs

34.6 Sample-size determination

- For means, proportions, correlations, and regression
- For survival outcomes
- For clustered designs (design effect)
- Software: G*Power, pwr (R), statsmodels (Python)

Chapter 35 Meta-Analysis and Research Synthesis

35.1 Foundations of meta-analysis

- Systematic review and meta-analysis
- Protocol, registration, and PRISMA
- Effect-size extraction and coding

35.2 Fixed- and random-effects models

- Inverse-variance weighting
- DerSimonian-Laird random effects
- Restricted maximum likelihood (REML) random effects
- Heterogeneity: Cochran's Q , I^2 , τ^2

35.3 Publication bias

- Funnel plot and its interpretation
- Egger's and Begg's tests
- Trim-and-fill method
- p-curve and z-curve analysis

35.4 Moderator analysis and meta-regression

- Subgroup analysis
- Meta-regression
- Bubble plots

35.5 Advanced methods

- Network meta-analysis
- Bayesian meta-analysis
- Individual patient data (IPD) meta-analysis
- Robust variance estimation (RVE)

Chapter 36 Spatial Statistics

36.1 Spatial data types

- Point data, areal data, and geostatistical data
- Coordinate systems and projections

36.2 Exploratory spatial analysis

- Choropleth maps and spatial visualization
- Spatial autocorrelation: Moran's I, Geary's C
- Local indicators: LISA, Getis-Ord G_i^*
- Join-count statistics

36.3 Point process models

- Complete spatial randomness (CSR)
- Ripley's K, L, and pair correlation functions
- Poisson and Cox processes

36.4 Geostatistics

- Variogram and covariance modeling
- Kriging (ordinary, universal, co-kriging)
- Spatial interpolation

36.5 Spatial regression

- Spatial lag model
- Spatial error model
- Geographically weighted regression (GWR)

Chapter 37 Circular and Directional Statistics

37.1 Circular data fundamentals

- Circular mean and variance

- Trigonometric moments
- The von Mises distribution
- Wrapped distributions

37.2 Inference for circular data

- Rayleigh test
- Watson's U^2 test
- Kuiper's test
- Rao's spacing test

37.3 Comparing circular distributions

- Watson-Williams test
- Mardia-Watson-Wheeler test
- Multi-sample tests

37.4 Spherical and higher-dimensional directional data

- Von Mises-Fisher distribution
- Spherical data visualization

Chapter 38 Extreme-Value Theory and Reliability

38.1 Foundations of EVT

- Fisher-Tippett-Gnedenko theorem
- Domains of attraction
- Tail index and heaviness

38.2 Block maxima approach

- Generalized extreme value (GEV) distribution
- Return levels and return periods
- Estimation: MLE, L-moments, Bayesian

38.3 Peaks-over-threshold approach

- Choosing the threshold
- Generalized Pareto distribution (GPD)
- Mean residual life plot
- Hill estimator

38.4 Reliability analysis

- Weibull life-data analysis
- Failure rate and bathtub curve
- MTBF and MTTF
- Accelerated life testing
- System reliability: series and parallel

Chapter 39 Psychometrics and Item Response Theory

39.1 Classical test theory

- True score model
- Reliability: Cronbach's alpha, McDonald's omega, KR-20
- Standard error of measurement
- Attenuation correction

39.2 Item response theory (IRT)

- Rasch model (1PL)
- 2PL and 3PL models
- Graded response and partial credit models
- Parameter estimation: MML, MCMC
- Person and item fit statistics

39.3 Scale development and validation

- Content validity
- Construct validity and structural validity
- Criterion validity
- Measurement invariance and DIF (Mantel-Haenszel DIF, LR-DIF)

39.4 Mokken scale analysis

- Non-parametric IRT
- Monotone homogeneity and double monotonicity

Chapter 40 Epidemiological Methods

40.1 Measures of disease frequency

- Incidence (risk and rate), prevalence, person-time
- Attack rate, secondary attack rate
- Mortality and case-fatality rates vs. infection-fatality rate

40.2 Measures of association and impact

- Risk difference, relative risk, odds ratio, hazard ratio
- Attributable fraction (exposed) and population attributable fraction
- Number needed to treat / harm

40.3 Standardization

- Direct and indirect age-standardization
- Standardized mortality / morbidity ratio (SMR)

40.4 Study designs

- Cohort, case-control (incl. nested, case-cohort), cross-sectional
- Ecological, case-crossover, randomized trial

40.5 Bias, confounding, and validity

- Selection bias (Berkson, healthy-worker, loss to follow-up)
- Information / misclassification bias (differential vs. non-differential)
- Confounding and effect modification
- Collider bias and collider stratification
- Immortal-time, lead-time, length-time bias

40.6 Diagnostic test evaluation

- Sensitivity, specificity, PPV, NPV, likelihood ratios
- Pre-test and post-test probability
- ROC / AUC analysis and DeLong's test
- Youden's J, optimal thresholds

40.7 Infectious disease metrics

- R_0 and R_t
- Herd-immunity threshold
- Serial interval, generation time, incubation period
- Compartmental models (SIR, SEIR, and extensions)

PART XIV

Machine Learning and Statistical Prediction

Chapter 41 Supervised Learning

41.1 The prediction framework

- Prediction vs. inference
- Bias-variance decomposition
- Training, validation, and test sets
- Loss functions (MSE, cross-entropy, etc.)

41.2 Tree-based methods

- Decision trees (CART)
- Bagging and random forests
- Gradient boosting (XGBoost, LightGBM, CatBoost)
- Variable importance measures

41.3 Support vector machines

- Hard and soft margin SVM
- Kernel trick
- SVM for regression (SVR)

41.4 Neural networks and deep learning

- Feedforward networks and backpropagation
- Convolutional neural networks (CNN)
- Recurrent networks (RNN, LSTM, GRU)
- Transformers
- Regularization: dropout, batch normalization

41.5 Model evaluation and selection

- Cross-validation (k-fold, nested, stratified)
- ROC/AUC, precision-recall, F1
- Calibration: Brier score, reliability diagrams
- Hyperparameter tuning: grid search, random search, Bayesian optimization

41.6 Interpretable and explainable ML

- SHAP values
- LIME
- Partial dependence plots
- Accumulated local effects

Chapter 42 Unsupervised Learning and Representation

42.1 Unsupervised vs. supervised learning

- Goals and evaluation challenges

42.2 Dimensionality reduction for ML

- PCA, ICA, autoencoders
- t-SNE and UMAP for visualization

42.3 Generative models

- Gaussian mixture models (also Chapter 20)
- Variational autoencoders (VAE)
- Generative adversarial networks (GAN)

42.4 Anomaly detection

- Statistical outlier methods
- Isolation forest
- One-class SVM
- Autoencoders for anomaly detection

PART XV

Data Transformation, Computation, and Workflow

Chapter 43 Data Transformations

43.1 Why transform?

- Stabilizing variance
- Achieving normality for parametric tests
- Linearizing relationships
- Interpretability

43.2 Power and log-family transforms

- Log transformation ($\ln$, $\log_{10}$)
- Square-root, reciprocal, and higher powers
- Tukey's ladder of powers
- Box-Cox transformation (positive data)
- Yeo-Johnson transformation (zero and negative values)
- Inverse hyperbolic sine (IHS / arcsinh)

43.3 Transforms for bounded data

- Arcsine-root transformation for proportions
- Logit transformation
- Folded transformations

43.4 Standardization and normalization

- Z-score standardization
- Min-max normalization
- Robust scaling (median and MAD)

43.5 Transforms for inference on derived quantities

- Fisher's z-transformation for correlation
- Rank-based inverse-normal (rankit / van der Waerden)
- Anscombe and Freeman-Tukey transforms for counts
- Delta method (also Chapter 7)

43.6 Transforms for time series

- Differencing for stationarity
- Seasonal adjustment
- Log and Box-Cox in ARIMA

Chapter 44 Computational Statistics and Simulation

44.1 Numerical optimization

- Newton-Raphson and quasi-Newton methods
- EM algorithm
- Gradient descent and stochastic gradient descent
- Coordinate descent

44.2 Monte Carlo methods

- Pseudorandom number generation
- Monte Carlo integration
- Importance sampling
- Variance-reduction techniques

44.3 Simulation for inference

- Simulation-based power analysis
- Parametric bootstrap via simulation
- Approximate Bayesian computation (ABC)

44.4 Statistical computing tools

- R: base stats, tidyverse, ggplot2, and key packages
- Python: NumPy, SciPy, statsmodels, scikit-learn, PyMC
- SAS and SPSS in applied practice
- Stan and JAGS for Bayesian computation

Chapter 45 Reproducibility, Reporting, and Research Practice

45.1 Reproducibility and replicability

- The replication crisis
- Computational reproducibility
- Preregistration and registered reports
- Open data and open code

45.2 Analytical pitfalls and inference traps

- Circular analysis / double dipping
- Data leakage and overfitting
- p-hacking and researcher degrees of freedom
- HARKing (hypothesizing after results known)
- Multiple comparisons and forking paths
- Regression to the mean
- Simpson's paradox
- Survivorship bias
- Ecological / atomistic fallacy
- Texas sharpshooter fallacy
- Base-rate neglect

45.3 Reporting standards

- APA, CONSORT, STROBE, PRISMA guidelines
- Reporting effect sizes and CIs
- Table and figure standards
- Supplementary materials and data sharing

45.4 Software and literate programming

- R Markdown and Quarto
- Jupyter notebooks
- Version control with Git
- Containerization (Docker) for reproducibility

PART XVII

Advanced Paradigms in Modern Data Science

Chapter 46 Functional Data Analysis (FDA)

Methods for data whose observations are functions, curves, or surfaces — e.g., time series, spectral profiles, anatomical tract profiles — where scalar-summary approaches discard distributional information. FDA extends classical multivariate methods to infinite-dimensional function spaces.

46.1 Foundations of functional data

- What makes data functional: observations as elements of a function space
- Discretely observed vs. continuously indexed functions
- Representing functional data: basis expansions (B-splines, Fourier, wavelets)
- Smoothing and regularisation: roughness penalties, cross-validated bandwidth
- Amplitude vs. phase variation

46.2 Functional descriptive statistics

- Functional mean, variance, and covariance surface
- Functional boxplot and outlier detection (modified band depth)
- Functional depth measures: Tukey, simplicial, h-mode depth
- Visualization: spaghetti plots, rainbow plots, functional boxplots

46.3 Functional principal component analysis (fPCA)

- Eigendecomposition of the covariance operator (Karhunen-Loève expansion)
- Choosing the number of functional principal components
- Scores as scalar summaries of each curve
- Bivariate fPCA and cross-covariance operators
- Robust fPCA

46.4 Functional regression models

- Scalar-on-function regression: scalar outcome, functional predictor
- Function-on-scalar regression: functional outcome, scalar predictors
- Function-on-function regression: both outcome and predictor are functional
- Concurrent (pointwise) model: $Y(t) \sim X(t)$ at each t
- Historical functional linear model: $Y(t)$ depends on $X(s)$ for $s \leq t$
- Estimation via penalised splines, fPCA basis, or kernel smoothing

46.5 Functional hypothesis testing

- Functional ANOVA (fANOVA): compare mean curves across groups
- Pointwise t-tests with simultaneous confidence bands (Bonferroni, bootstrap)
- Globaltest and L2-norm-based omnibus tests
- Permutation-based tests for functional data

- Testing equality of covariance operators across groups

46.6 Registration and curve alignment

- Phase variation and the need for alignment before amplitude analysis
- Landmark registration: align curves at anatomically defined features
- Continuous registration: minimise a warping penalty (Fisher-Rao metric)
- Dynamic time warping (DTW) as a special case
- Square-root velocity framework for elastic curves

46.7 Functional clustering and classification

- k-means clustering on fPC scores
- Model-based functional clustering (mixture of Gaussian processes)
- Functional discriminant analysis
- Functional random forests and support vector machines

46.8 Software and applications

- R: fda (Ramsay & Silverman), fdapace, refund, roahd
- Python: scikit-fda
- Applications: tract-profile diffusion MRI, relaxometry curves, EEG/fMRI time series, growth curves, spectroscopy
- Connection to longitudinal data analysis (Chapter 25) and time series (Chapters 26-27)

Chapter 47 Compositional Data Analysis (CoDa)

Statistical methods for data constrained to a simplex — vectors of positive parts that sum to a constant (e.g., tissue fractions, species proportions, budget shares). Standard Euclidean methods are invalid in simplex geometry; log-ratio transforms map compositions to unconstrained space where classical statistics apply.

47.1 The simplex and the problem with Euclidean statistics

- Definition of a composition: parts summing to a constant (D-part simplex S^D)
- The spurious correlation problem: dividing by a total induces negative correlations
- Subcompositional coherence: conclusions about a subset of parts must not depend on excluded parts
- Scale invariance: only ratios between parts carry information
- Perturbation and powering: the algebraic operations on the simplex

47.2 Log-ratio transformations

- Additive log-ratio (ALR): $\log(x_i / x_D)$ for a reference part D — simple but asymmetric
- Centred log-ratio (CLR): $\log(x_i / g(x))$ where $g(x)$ is the geometric mean — symmetric but singular
- Isometric log-ratio (ILR): orthonormal coordinates via sequential binary partition — fully Euclidean, preferred for regression and distance analysis
- Pivot coordinates: a useful ILR variant where each coordinate contrasts one part against the remaining
- Choosing among transforms: ALR for simple two-part contrasts, CLR for PCA and visualisation, ILR for regression

47.3 Aitchison geometry and distances

- Aitchison inner product, norm, and distance on the simplex

- Aitchison distance = Euclidean distance in CLR space
- Subcompositional dominance property
- Comparison with other compositional distances (Bray-Curtis, Hellinger)

47.4 Exploratory analysis of compositions

- CoDa biplot in CLR space: rays show parts, vertices show observations
- Variation matrix: $\text{var}(\log(x_i/x_j))$ for all pairs — displays which ratios vary most
- Ternary diagrams for three-part compositions
- Compositional PCA (clr-PCA): PCA on CLR-transformed data

47.5 Regression with compositional data

- Compositional predictor, scalar outcome: apply ILR, then standard regression
- Scalar predictor, compositional outcome: Dirichlet regression or ILR-multivariate regression
- Compositional predictor and outcome: ILR on both sides
- Interpretation: regression coefficients relate to log-ratio contrasts, not absolute part sizes
- Linear models in the simplex: perturbation-regression framework

47.6 Zeros and missing data in compositions

- Structural zeros vs. rounded zeros (below detection limit)
- Multiplicative zero replacement (Martin-Fernandez et al.)
- Bayesian imputation approaches for zeros and missing parts
- Half-detection-limit imputation: when appropriate and when not

47.7 Testing and inference for compositions

- Permutation MANOVA in Aitchison geometry (PERMANOVA with Aitchison distance)
- Compositional ANOVA via ILR + standard F-tests
- Testing subcompositional independence
- Bootstrap confidence regions on the simplex

47.8 Software and applications

- R: compositions, robCompositions, zCompositions, DirichletReg
- Python: scikit-bio (skbio.stats.composition)
- Applications: tissue volume fractions in neuroimaging, microbiome abundance data, geochemistry, diet analysis, economic budget shares
- Connection to Dirichlet distribution (Chapter 3) and multivariate analysis (Chapters 18-19)

Chapter 48 Topological Data Analysis (TDA)

Methods from algebraic topology — persistent homology, the Mapper algorithm, and related tools — applied to data. TDA characterises the shape and multi-scale structure of point clouds, networks, and function spaces in a coordinate-free, noise-robust way that complements distance- and covariance-based methods.

48.1 Motivating ideas: shape and topology

- Topology studies properties invariant under continuous deformation: connectivity, loops, voids
- Why topology: scale-free, threshold-free characterisation of structure in high-dimensional data
- Betti numbers: β_0 (connected components), β_1 (loops/cycles), β_2 (voids/cavities)

- Euler characteristic: alternating sum of Betti numbers — a scalar topological summary

48.2 Simplicial complexes and filtrations

- Simplicial complex: a set of vertices, edges, triangles, and higher-dimensional simplices
- Vietoris-Rips complex: connect points within radius r ; Čech complex: require a common ball
- Alpha complex: sub-complex of the Delaunay triangulation — more efficient for Euclidean data
- Filtration: a nested sequence of complexes parameterised by a scale parameter
- Function-induced filtrations (sub-level set filtrations) for scalar fields

48.3 Persistent homology

- Track the birth and death of topological features across the filtration
- Persistence diagram: scatter plot of (birth, death) pairs; points near diagonal are noise
- Barcode: alternative representation with horizontal intervals [birth, death]
- Persistence = death - birth: measure of feature significance
- Stability theorem: small perturbations in data produce small changes in the diagram (bottleneck distance)

48.4 Distances and statistics on persistence diagrams

- Bottleneck distance: minimax matching cost between two diagrams
- p-Wasserstein distance: average matching cost ($p=2$ most common)
- Permutation tests using Wasserstein distance for two-sample comparison
- Confidence sets for persistence diagrams via bootstrapping
- Mean persistence diagram (Frechet mean in Wasserstein space)

48.5 Vectorisation: persistence diagrams as ML features

- Persistence landscape: piecewise linear function summarising the diagram
- Persistence image: pixelated 2D representation with Gaussian smoothing (Adams et al., 2017)
- Betti curves and Euler characteristic curve: scalar functions over filtration parameter
- Persistent entropy: entropy of the persistence lifetime distribution
- Using these features in standard ML pipelines (SVM, random forest, neural networks)

48.6 The Mapper algorithm

- Goal: construct a compressed graph representing the topology of high-dimensional data
- Three ingredients: a filter function (e.g., PCA score, density estimate), a cover of the filter range, and a clustering algorithm
- Nerve of a cover: vertices are clusters, edges connect overlapping clusters
- Parameter sensitivity and how to choose cover resolution and overlap
- Software: KeplerMapper (Python), TDAmapper (R)

48.7 Applications and software

- Neuroscience: functional connectivity networks, white matter topology, brain network organisation
- Genomics and single-cell: cell-type trajectory analysis, cancer subtype identification
- Materials science: microstructure characterisation
- R: TDA, TDAmapper, ripserr
- Python: Ripser, Gudhi, giotto-tda, persim, scikit-tda
- Computational considerations: Ripser is highly optimised; Gudhi supports alpha complexes

Chapter 49 Joint Longitudinal–Survival Models

Models that simultaneously specify a mixed-effects model for a longitudinal biomarker and a survival model for a time-to-event outcome, linked through shared latent quantities. They correct for informative dropout (patients who experience the event are no longer observed) and provide combined inference unavailable from separate analyses.

49.1 Motivation and the problem with two-stage analyses

- The informative dropout problem: patients who die or convert cannot be observed longitudinally
- Why the naive two-stage approach (plug predicted values into Cox model) is biased
- The measurement error problem: observed biomarker values contain noise; Cox models need the true value
- Overview of joint modelling as the unified solution

49.2 The joint model framework

- Longitudinal sub-model: linear mixed-effects model for the biomarker trajectory (Chapter 17)
- Survival sub-model: Cox proportional-hazards model for the event time (Chapter 24)
- Association structure: how the longitudinal process enters the hazard
- Shared random effects as the linking mechanism
- The full joint likelihood: product of the marginal longitudinal and conditional survival likelihoods

49.3 Association structures

- Current value: hazard depends on the true biomarker value at the event time — most common
- Current slope: hazard depends on the rate of change of the trajectory
- Cumulative effect (integral): hazard depends on the area under the biomarker trajectory
- Time-lagged effects: hazard at time t depends on trajectory at time $t - \text{lag}$
- Choosing association structure: subject-matter knowledge and model comparison (WAIC, LOO-CV)

49.4 Estimation

- Maximum likelihood via the EM algorithm or direct numerical optimisation
- Gaussian-Hermite quadrature to integrate over the random-effects distribution
- Bayesian estimation: full posterior via MCMC in Stan or JAGS
- R packages: JM (Rizopoulos), JMBayes2 (Bayesian and frequentist), rstanarm joint models

49.5 Extensions

- Multivariate joint models: multiple longitudinal outcomes with one survival outcome
- Multi-state joint models: intermediate events (e.g., remission then relapse)
- Competing risks in the survival sub-model
- Non-linear longitudinal trajectories: splines and polynomial growth curves in the mixed model
- Recurrent event outcomes: gap-time and intensity models

49.6 Dynamic predictions

- At each visit, update the individual survival prediction using all observed biomarker history
- Posterior survival probabilities conditional on observed longitudinal data up to current time
- Calibration and discrimination of dynamic predictions (AUC, Brier score at horizon t)
- Clinical application: personalised monitoring and treatment decision rules

- Software: JMbayes2::dynInfo and survfitJM; timeROC for evaluation

49.7 Connections and related models

- Two-stage approach: show why it fails and when it approximately holds
- Landmark analysis: simpler alternative — use biomarker history up to landmark time as predictors
- Pattern-mixture models and selection models: alternative frameworks for informative dropout (Chapter 33)
- Growth-curve models (Chapter 17) and competing risks (Chapter 24) as building blocks
- Connection to joint modelling of longitudinal and recurrent events

Chapter 50 Graphical Models and Causal Discovery

Probabilistic graphical models encode conditional independence structure as a graph. Causal discovery algorithms learn that graph from data rather than specifying it a priori — a goal distinct from the causal identification methods in Chapter 23.

50.1 Undirected graphical models (Markov random fields)

- Conditional independence and Markov properties: global, local, and pairwise
- Gaussian graphical model (GGM): precision matrix encodes the conditional independence graph
- Ising model for binary variables; Potts model for categorical variables
- Estimation: graphical Lasso (glasso), CLIME, neighbourhood selection (Meinshausen-Buhlmann)
- Model selection: StARS, cross-validation, BIC; R: huge, glasso; Python: sklearn

50.2 Directed graphical models and Bayesian networks

- DAGs, d-separation, and the Markov condition
- Markov equivalence classes (MECs) and CPDAGs
- Faithfulness assumption and its empirical testability
- Score-based learning: BIC, BDeu; GES algorithm
- Constraint-based learning: PC algorithm, FCI (with hidden confounders)

50.3 Causal discovery algorithms

- PC algorithm: CI tests, skeleton estimation, V-structure orientation, orientation rules
- FCI: extension of PC to allow latent variables; outputs partial ancestral graph (PAG)
- LiNGAM: exploit non-Gaussianity to fully identify DAG beyond the MEC
- NOTEARS: continuous optimisation for DAG structure via acyclicity constraint on matrix exponential
- Score-based deep learning methods: DAGS-GNN, NoCurl

50.4 Exponential random graph models (ERGMs)

- Model probability distribution over networks as exponential family in graph statistics
- Sufficient statistics: edges, triangles, degree sequence, k-stars
- Estimation via MCMC-MLE; degeneracy problem and near-degenerate parameter regions
- Goodness-of-fit via simulated network diagnostic plots
- R: ergm; Python: graph-tool

50.5 Stochastic block models and latent network models

- Stochastic block model (SBM): nodes in blocks; block-specific edge probabilities

- Degree-corrected SBM: accounts for heterogeneous degree sequences
- Community detection: spectral clustering, variational EM, belief propagation
- Latent space model (Hoff et al.): edge probability as function of latent node positions
- Random dot product graph: inner product of latent vectors drives edge probability

50.6 Software and connections

- R: pcalg, bnlearn, glasso, huge, ergm, latentnet, igraph
- Python: pgmpy, gcastle, networkx, grakel, graph-tool
- Connections: causal inference (Chapter 23), TDA (Chapter 48), dimension reduction (Chapter 19)

Chapter 51 Post-Selection and High-Dimensional Inference

Data-adaptive model selection (Lasso, stepwise, cross-validation) invalidates standard inference. Post-selection inference restores Type I error control. High-dimensional methods address the $p \gg n$ regime.

51.1 The post-selection inference problem

- Selection bias: standard CIs and p-values are anti-conservative after data-driven selection
- Selective Type I error: condition on the selection event to define the relevant error rate
- Truncated-normal pivot for Lasso via the polyhedral lemma (Lee et al. 2016)
- Data splitting: half for selection, half for inference; universally valid for any selector
- PoSI: simultaneous post-selection inference valid for any possible model chosen

51.2 Lasso inference

- Debiased (de-sparsified) Lasso: add score-correction to restore $\sqrt{n}$ -normal limiting distribution
- Confidence intervals for individual Lasso coefficients (Zhang & Zhang; van de Geer et al.)
- SLOPE: sorted L1 penalty achieving simultaneous FDR control among regression coefficients
- Stability selection: resampling-based selection frequency as measure of variable importance
- R: hdi, SLOPE; Python: statsmodels, sklearn

51.3 Knockoffs and FDR-controlled variable selection

- Fixed-X knockoffs (Barber & Candès 2015): knockoff matrix enables finite-sample FDR control
- Model-X knockoffs (Candès et al. 2018): knockoffs sampled from known feature distribution
- Knockoff filter statistic: any measure of relative importance of variable vs its knockoff
- Deep knockoffs: learn knockoff distribution when feature distribution is complex
- Applications: GWAS, neuroimaging feature selection, drug discovery

51.4 Global tests and high-dimensional aggregation

- Higher Criticism (HC, Donoho & Jin 2004): optimal detection of sparse mixtures in noise
- Berk-Jones test: supremum test on empirical vs null CDF; more powerful than HC in some regimes
- SKAT and SKAT-O: kernel association tests for gene sets and pathways in GWAS
- Min-p and Tippett test: global null test via minimum p-value across hypotheses

51.5 Random matrix theory for high-dimensional settings

- Marchenko-Pastur law: eigenvalue bulk distribution when $p/n \rightarrow c$
- Tracy-Widom distribution: fluctuations of the largest eigenvalue under the null

- Spiked covariance model: signal eigenvalues above the bulk vs noise
- Ledoit-Wolf and Oracle Approximating Shrinkage: optimal linear shrinkage of sample covariance
- Concentration of measure: pairwise distances and inner products concentrate as dimension grows - the geometric face of the curse of dimensionality
- R: RMTstat, corpcor; Python: sklearn

Chapter 52 Conformal Prediction and Distribution-Free Uncertainty

Conformal prediction constructs prediction sets with finite-sample, assumption-free coverage guarantees for any black-box predictive model, requiring only exchangeability. No distributional assumptions needed.

52.1 Foundations and split conformal

- Nonconformity score: measure of how unusual a test point is relative to the training set
- Split conformal: calibration set quantile of nonconformity scores defines the prediction interval
- Coverage guarantee: $P(Y_{\text{test}} \in C(X_{\text{test}})) \geq 1 - \alpha$, exactly, for exchangeable data
- Full conformal: refit for each candidate label; computationally intensive but uses all data
- R: MAPIE, cfcplib; Python: MAPIE, crepes, nonconformist

52.2 Efficiency improvements

- Cross-conformal / CV+: K-fold variants with marginal coverage; better efficiency than split
- Jackknife+ (Barber et al. 2021): LOO prediction intervals with valid marginal coverage
- Mondrian conformal: stratified by class; guarantees conditional coverage within strata
- Locally weighted conformal: reweight scores by density ratio; adapts to local difficulty
- Weighted conformal: importance weighting for covariate shift

52.3 Generalised risk control

- Conformal risk control (CRC, Angelopoulos et al. 2022): control any monotone risk function at target level
- Risk-controlling prediction sets (RCPS, Bates et al. 2021): high-probability control of a monotone risk
- Learn then Test (LTT, Angelopoulos et al. 2022): multiple testing for risk parameter selection
- Applications: medical imaging (FNR control), NLP (hallucination rate), drug discovery

52.4 Prediction-powered inference

- Setting: large unlabelled ML predictions + small labelled ground-truth dataset
- PPI estimator: augmented estimator using prediction residuals; valid CIs with boosted power
- PPI++ (Angelopoulos et al. 2023): optimal weighting of labelled and prediction contributions
- Applications: protein property estimation, ecological surveys, census augmentation
- Python: ppi_py; no distributional assumptions required

Chapter 53 E-Values and Safe Anytime-Valid Inference

E-values are non-negative statistics with expected value ≤ 1 under the null, interpretable as wealth when betting against the null. They are valid under optional stopping and compose multiplicatively, enabling safe sequential inference.

53.1 E-values: foundations

- Definition: E is an e-value if $E_{H_0}[E] \leq 1$; measures evidence as bettor's wealth

- Bayes factors are e-values; $1/E$ is a valid p-value by Markov inequality
- p-to-e calibration: any p-value can be converted to an e-value
- Likelihood ratios as e-values: Neyman-Pearson framework in e-value language

53.2 E-processes and sequential validity

- Test martingale: product of e-values forms a non-negative supermartingale under H_0
- Ville's inequality: $P(\sup_t E_t \geq 1/\alpha) \leq \alpha$; anytime-valid Type I error control
- Optional stopping: valid to stop whenever the e-process exceeds a threshold
- Anytime-valid t-test: valid at any sample size without alpha-spending
- R: safestats; Python: anytime-valid

53.3 Merging and combining e-values

- Arithmetic mean: valid merger for independent e-values
- Harmonic mean: robust to heavy tails; valid for e-values with finite variance
- e-BH procedure: BH on e-values controls FDR under arbitrary dependence
- Universal inference: split likelihood ratio is an e-value for composite and non-parametric nulls

53.4 Safe tests and the GROW criterion

- GROW criterion: maximise expected log e-value (= expected information) under the alternative
- Safe tests unify SPRT, likelihood ratio tests, and Bayes factors under one framework
- Information-theoretic interpretation: e-value = $\exp(\text{per-observation KL contribution})$
- Connections to conformal prediction (Chapter 52) and Bayesian inference (Chapter 28)

Chapter 54 Causal Machine Learning

Causal ML combines semi-parametric efficiency theory with flexible ML to estimate average and individualised treatment effects with valid inference in high-dimensional settings.

54.1 Double/debiased machine learning (DML)

- Partial linear model: $Y = \theta D + g(X) + \epsilon$; estimate θ despite high-dimensional g
- Neyman orthogonality: score function has zero first derivative w.r.t. nuisance at the truth
- Cross-fitting: K-fold procedure that removes regularisation bias; preserves $\sqrt{n}$ normality
- Extensions: average treatment effect, LATE, quantile treatment effects
- R: DoubleML; Python: doubleml, econml

54.2 Doubly-robust and efficient estimators

- AIPW: augmented IPW; consistent if either propensity or outcome model is correct
- TMLE: targeted one-step correction achieves semi-parametric efficiency bound
- DR-learner: doubly-robust CATE estimator using doubly-robust pseudo-outcomes
- Semi-parametric efficiency bound: minimum asymptotic variance for any regular estimator of ATE
- R: AIPW, tmle; Python: econml, zEpid

54.3 Heterogeneous treatment effect estimation

- CATE: $\tau(x) = E[Y(1) - Y(0) | X=x]$; varies across individuals
- Causal forest (Wager & Athey 2018): honest splitting; valid pointwise CIs for CATE

- R-learner: regress outcome residuals on treatment residuals; flexible base learner
- X-learner: cross-fitted imputation of individual effects; effective for unequal group sizes
- Meta-learner taxonomy (S-, T-, X-, R-, DR-): R: grf, rlearner; Python: econml

54.4 DiD extensions and synthetic controls

- Staggered DiD: units treated at different times; Callaway-Sant'Anna, Sun-Abraham estimators
- Synthetic DiD (SDID): combines SC weighting with DiD; robust to parallel trends violations
- Interactive fixed effects: absorb time-varying confounders via latent factor structure
- Triple difference (DDD); semiparametric propensity-score-weighted DiD (Abadie 2005)
- R: did, synthdid, fixest; Python: linearmodels

54.5 Policy learning

- Empirical welfare maximisation: choose policy maximising doubly-robust value estimate
- Policy trees: interpretable rectangular decision rules; regression tree on pseudo-outcomes
- Regret bounds and sample complexity for policy learning
- R: policytree; Python: econml

Chapter 55 Bayesian Nonparametrics

BNP models place priors over infinite-dimensional objects (distributions, functions, partitions) so that complexity grows with data. The Dirichlet process and its extensions are the core building blocks.

55.1 Dirichlet process

- $DP(\alpha, G_0)$: prior over distributions; realisations discrete a.s.; base measure G_0 , concentration α
- Stick-breaking construction: $\pi_k = V_k * \prod_{j < k} (1 - V_j)$; $V_k \sim \text{Beta}(1, \alpha)$
- Chinese restaurant process: sequential partition interpretation; $E[\text{clusters}] \sim \alpha * \log(n)$
- Posterior: $DP(\alpha + n, (\alpha * G_0 + \sum \delta_{x_i}) / (\alpha + n))$; self-conjugate

55.2 DP mixtures and inference

- DPM: $Y_i | \theta_i \sim F(\theta_i)$; $\theta_i | G \sim G$; $G \sim DP$; number of components inferred
- Collapsed Gibbs sampler (Neal 2000): integrate out G ; sample cluster assignments directly
- Split-merge MCMC: improve mixing; slice sampling for large datasets
- R: DPpackage, PReMiuM; Python: BNPy, PyMC

55.3 Extensions: Pitman-Yor, HDP, IBP

- Pitman-Yor process: discount d in $[0, 1)$; power-law cluster sizes; foundation of PY language models
- Hierarchical DP: share atoms across groups; HDP-HMM for infinite-state HMMs
- Indian buffet process: prior over binary feature matrices with unbounded columns
- Beta process: completely random measure; de Finetti measure for IBP; sparse coding priors

55.4 Computation and software

- Truncated stick-breaking: finite approximation to DP; variational EM
- Variational BNP: mean-field VI with stick-breaking truncation; scalable
- Particle MCMC for latent feature models
- R: DPpackage, PReMiuM, bnptools; Python: BNPy, Pyro, NumPyro

Chapter 56 Simulation-Based Inference and Likelihood-Free Methods

When the likelihood is intractable but data can be simulated, neural estimators approximate the posterior, likelihood, or likelihood ratio directly from simulator output, enabling Bayesian inference for complex models.

56.1 Approximate Bayesian computation (ABC)

- Rejection ABC: accept θ if $\text{dist}(\text{sim_stats}(\theta), \text{obs_stats}) < \epsilon$
- ABC-MCMC and ABC-SMC: more efficient sampling; sequential Monte Carlo for multi-modal posteriors
- Summary statistics: sufficiency is optimal; neural network learned summaries (Jiang et al.)
- R: `abc`, `EasyABC`; Python: `pyabc`, `abcpyp`; ϵ controls approximation-acceptance trade-off

56.2 Neural SBI: SNPE, NLE, NRE

- SNPE: normalising flow trained to approximate $p(\theta | x)$; active learning rounds reduce simulations
- NLE: conditional density estimator for $p(x | \theta)$; combine with MCMC for posterior
- NRE: classifier distinguishing (x, θ) from joint vs marginal; ratio = likelihood ratio
- Python: `sbi`; benchmark: `sbibm`; choose method based on simulation budget and posterior shape

56.3 Normalising flows

- Invertible neural networks: map simple base to complex target distribution; tractable density
- Variants: RealNVP, MAF (Masked Autoregressive Flow), NSF (Neural Spline Flow), FFJORD
- Training: maximum likelihood on samples; Python: `nflows`, `zuko`, `normflows`
- Applications: posterior approximation in SBI, variational inference, generative modelling

56.4 Score-based methods and diffusion

- Score function: $\nabla_x \log p(x)$; estimate via denoising score matching (Hyvarinen 2005)
- Langevin dynamics: sample from $p(x)$ via $x \leftarrow x + \epsilon \cdot \text{score}(x) + N(0, 2\epsilon)$
- DDPM (Ho et al. 2020): forward Gaussian noising + learned neural network reverse
- Score SDE framework (Song et al. 2021): continuous-time diffusion; ODE and SDE samplers

56.5 Variational inference and amortisation

- ELBO: evidence lower bound; maximise to train VAE; gap = $\text{KL}(q||p_{\text{posterior}})$
- IWAE: importance-weighted ELBO; tighter bound using multiple importance samples
- Amortised inference: encoder network maps x to posterior parameters; no MCMC at test time
- R: `rstan` (ADVI); Python: `Pyro`, `NumPyro`, `blackjax`, `flowjax`

Chapter 57 Dynamic Treatment Regimes and Reinforcement Learning

Dynamic treatment regimes formalise adaptive sequential decisions in medicine and policy. Reinforcement learning provides the mathematical and algorithmic framework for learning optimal regimes from data.

57.1 Dynamic treatment regimes

- DTR: sequence of decision rules $d_t(H_t) \rightarrow A_t$; evaluated by expected long-term outcome $E[Y^d]$
- Optimal DTR d^* : maximises $E[Y^d]$; estimated from observational or experimental data

- Structural nested mean models (SNMMs): semi-parametric framework for blip functions (causal contrasts)
- SMART design: randomise at each stage; provides data for efficient DTR estimation
- R: DTR, DynTxRegime

57.2 Q-learning and A-learning

- Q-function: $Q_t(h, a) = E[Y \mid H_t = h, A_t = a, \text{follow } d^* \text{ thereafter}]$
- Q-learning: backward induction via regression at each stage; simple but sensitive to model misspecification
- A-learning: model only the treatment contrast (advantage); doubly-robust; more efficient
- Weighted Q-learning: inverse-probability weights for non-randomised data

57.3 Off-policy evaluation and offline RL

- IPS estimator: reweight outcomes by importance ratio $\pi_{\text{target}}(a|s) / \pi_{\text{behaviour}}(a|s)$
- Doubly-robust OPE: augment IPS with direct method; consistent if either component is correct
- Offline RL: learn policy from fixed dataset; conservative Q-learning avoids distributional shift
- Python: d3rlpy, COBS; connection to causal inference (Chapter 23) and DTR

57.4 MDP and reinforcement learning fundamentals

- MDP: states S , actions A , transitions $P(s'|s, a)$, rewards $R(s, a)$, discount γ
- Bellman equations: $V(s) = \max_a [R(s, a) + \gamma \sum_{s'} P(s'|s, a) V(s')]$
- Model-based RL: estimate transition model, plan via value iteration
- Model-free RL: Q-learning (off-policy), SARSA (on-policy); TD learning

57.5 Policy gradient and contextual bandits

- Policy gradient (REINFORCE, PPO, A3C): directly optimise parameterised policy via gradient ascent
- Actor-critic methods: combine value function and policy gradient for variance reduction
- Contextual bandit: single-stage; Thompson sampling, UCB, epsilon-greedy
- R: contextual; Python: stable-baselines3, vowpalwabbit; connects to sequential and adaptive designs (Section 6.7)

PART XVI

Appendices

Chapter A Mathematical Notation and Symbols

A.1 Standard notation

- Greek letters used as parameters
- Roman letters as statistics
- Common operators and functions

A.2 Matrix algebra review

- Vectors, matrices, determinants, inverses
- Eigendecomposition and SVD
- Kronecker product and vec operator

Chapter B Statistical Tables

B.1 Standard tables

- Standard normal (z) table
- Student's t critical values
- Chi-squared critical values
- F critical values

B.2 Selected non-parametric tables

- Mann-Whitney U critical values
- Wilcoxon signed-rank critical values
- Spearman and Kendall critical values

Chapter C Probability Distribution Reference

C.1 Discrete distributions — summary

- PMF, mean, variance, MGF for each family

C.2 Continuous distributions — summary

- PDF, CDF, mean, variance, MGF for each family

Chapter D Test and Method Decision Guide

D.1 Choosing a test: decision flowchart

- Step 1: What is the outcome type?
- Step 2: How many groups or predictors?
- Step 3: Are observations independent or paired/repeated?
- Step 4: Are distributional assumptions met?
- Step 5: What is the goal — test, estimate, predict, or describe?

D.2 Assumption violations and fixes

- Non-normality
- Heteroscedasticity
- Outliers and influence
- Multicollinearity
- Non-independence

Chapter E Glossary**E.1 Key terms A–M**

- Full glossary of statistical terms

E.2 Key terms N–Z

- Continuation of glossary

Chapter F Cross-Reference: Tests and Methods by Application**F.1 Indexed by discipline**

- Biomedical and clinical research
- Psychology and education
- Economics and finance
- Environmental and ecological science
- Engineering and quality
- Social sciences

F.2 Indexed by software function

- R function cross-reference
- Python (SciPy / statsmodels / sklearn) cross-reference

Scope Notes

Breadth. This outline aims to be inclusive rather than selective — it covers the inferential statistics tradition (Parts III–IX), the modeling tradition (Parts VI–VIII), the machine learning tradition (Part XIV), and the applied domains (Parts XIII, X). Some items near the boundaries of statistics and other disciplines

(deep learning architectures, spatial databases, clinical trial regulation) are included as topics rather than developed at length.

Depth calibration. Not every bullet needs equal treatment. Foundational chapters (Parts I–III) should be written at introductory depth with no assumed background. Later parts progressively assume earlier material. Appendices are reference-level, not tutorial.

What is deliberately excluded. Proofs that belong in a mathematical statistics rather than applied text, specialized software documentation, and domain-specific clinical/regulatory standards (ICH E9, FDA guidance). Those can appear in footnotes or appendices.